

```
void vunmap_pmd_range(pmd, addr, end) {  
    pmd_t *pmd = pmd_offset(pud, addr);  
    do {  
        ...  
        // try remove huge page entry  
        int cleared = pmd_clear_huge(pmd);  
        ...  
        if (pmd_none_or_clear_bad(pmd))  
            continue;  
        ...  
        // try remove underlying PTEs  
        vunmap_pte_range(pmd, addr, next, mask);  
        ...  
    }  
    while (pmd++, addr = next, addr != end);  
} /* mm/vmalloc.c */
```

Find a 2MB entry (PMD) from the addr. of a 1GB entry (PUD)

Overloaded semantics:

- (1) The entry does not point to a 2MB huge page
- (2) The entry is not a directory
- (3) No entries of 4KB pages in the 2MB address range

The iterator assumes the spatial locality of entries